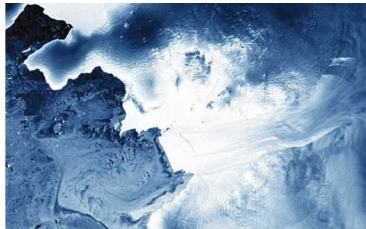


Small Satellites: All Set To Spur Innovation

Space is all set to be invaded by microsatellites, for developmental and commercial intents. These little satellites could help understand weather phenomena, ensure up-to-date maps, aid disaster relief, and more; but their curious cameras would also know when somebody remodels their house, or flies their private jet



▲ A Planet Labs microsat



◀ Using data from satellites, scientists found that from 2007 to 2008, six cubic kilometres of water drained from a lake in Antarctica—the largest drainage ever recorded. The amount of water is about one-tenth of the melting that occurs beneath the Antarctic surface each year (Courtesy: Planet Labs)

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Indian universities developed and launched several microsatellites (microsats) between 2009 and 2011, in association with the Indian Space Research Organisation (ISRO). These were amongst the first few microsats in space. While these were successful experiments that gave wings to academic dreams, microsats are now being viewed with greater expectations. These mini space wonders might work their magic beyond the field of science and technology and shake up the business world as well, much on the lines of GPS and location-based services.

"The term microsatellite is generally used for satellites in the range of 10 to 100 kg. Today, satellites generally weigh 1000 to 5000 kg," explains D. John, former deputy director, Satellite Communication Programme and deputy programme director, EDUSAT, ISRO headquarters, Bengaluru. Microsatellites are not only low in weight, but are also smaller and cost less than regular satellites. This enables them to be sent into space using launch vehicles or by piggybacking on rockets with excess capacity. Plus, due to their comparative simplicity, it also becomes possible to build and launch a fleet or constellation of such satellites, which might be used jointly as a communica-

tion or imagery system. Such a system will be capable of grabbing and updating images much faster than traditional satellites, due to the large number of satellites per constellation.

Imagine such constellations of microsats, launched and maintained by private companies, and leased out to businesses or research organisations. It would be possible to know in near real-time about events related to maritime movements, border security, forest fires, migratory birds, traffic situations, diversions, the status of crops, weather changes, map updates, or even the movement of a fleet of trucks, without requiring RFID tags or any cumbersome set-ups!